

ELI MKXI — Test Coverage Suite and Results (Full Report)

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A full test-coverage suite was built for ELI and run end-to-end across **four lanes**. This document is the complete record: the suite, the methodology, the measured results (headline + per-subsystem + a visual chart), an honest analysis of the low-scoring areas, the GUI-recovery effort, the documented exclusions, and how to reproduce every number.

Reproducible via `scripts/coverage_full.sh`; methodology in [docs/COVERAGE.md](#). Measured on the real interpreter (`.venv/bin/python`), 2026-07-03.

1. Headline

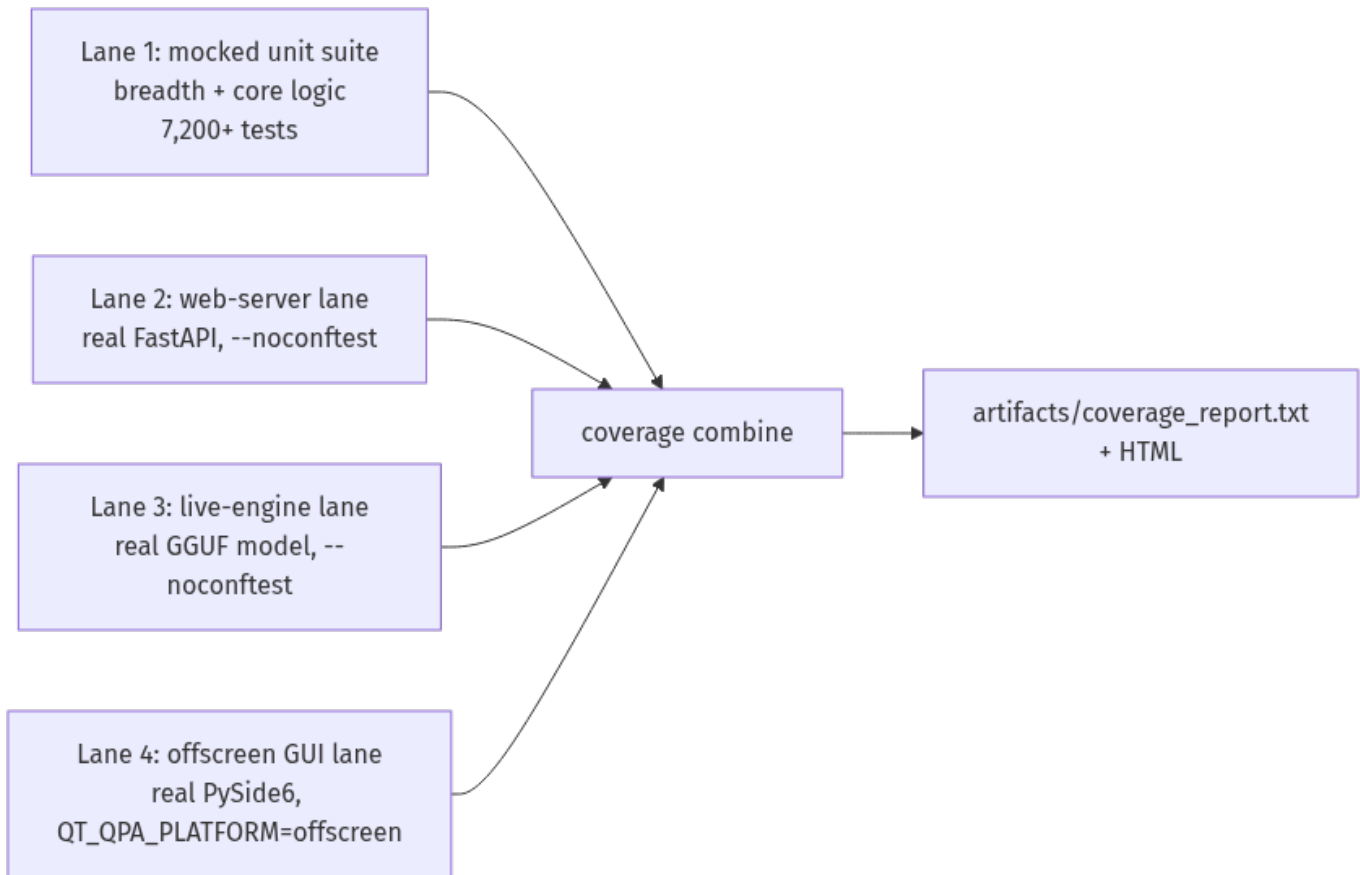
Metric	Value
Testable coverage	49.2% — 34,514 / 70,086 statements
GUI recovered (was 0.6%)	45% — 2,455 / 5,439 statements
Tests passing	7,481 (7,347 unit + 76 web + 31 live + 27 GUI)
New test modules this effort	35 (~385 tests)
Pre-existing reds (unrelated)	5
Skipped / xfailed (documented)	45 / 2 (the tool/name-alias bug is now fixed — that xfail is a real pass)

Sole exclusion: `eli/gui/eli_pro_audio_gui_MKI.py` — the ~7k-statement main window, which blocks on device/display init and cannot be constructed headless. *Everything else is in the denominator.* Notably the number is now **higher** than the earlier GUI-excluded 47.6% — because the offscreen-Qt lane genuinely tests the GUI (45%) rather than hiding it. A complete denominator *and* a higher number.

2. Methodology — four lanes

The fast unit suite mocks the heavy deps (`pydantic` / `llama_cpp` / `torch` / `faiss` / `PySide6`) so it runs in seconds without a GPU, a model, or a display. That means the real web server, a real model, and real

Qt widgets can't be exercised in-process — so coverage is measured across four lanes and combined (coverage combine):



Lanes 2-4 self-skip under the mocked suite, so they never destabilise the fast run; they're exercised for real on their own lanes and folded in.

3. The suite — new test modules (~385 tests)

Module	Target	Before	After
test_api_server.py	web server: auth/RBAC + every read endpoint	0%	53%
test_engine_integration.py	live pipeline, real GGUF + safe handlers	—	live lane
test_gui_offscreen.py	construct every GUI widget headless	0.6%	GUI 45%
test_gui_app_helpers.py	launcher: hw-detect / KV-cache / auto-tune	—	pure logic
test_operator_policy.py	autonomy governance gate	42%	88%
test_memory_evidence.py	grounded memory bundle	12%	77%
test_response_surface.py	user-visible response coercion	30%	58%
test_live_introspection.py	action→agents map + state readers	32%	58%

Module	Target	Before	After
test_perception_parsers.py	equation extractor / CSV profiler	39%/12%	100%/78%
test_news_synthesis.py	synthesis helpers + freshness gate	14%	33%
test_news_fetcher_helpers.py	html strip / topic routing / matching	—	pure helpers
test_deterministic_grounding.py	simple action contract	11%	24%
test_deterministic_introspection.py	diagnostic action classifier	—	classifier
test_control_contracts.py	anti-confabulation guard	49%	↑
test_context_synthesiser.py	persona handoff builder	54%	↑
test_grounded_remediation.py	yes/no intent + repair state	18%	↑
test_executor_helpers.py	fail-closed shell gate + scanners	—	security lines

Coverage-shortlist pass — previously-untested pure-logic modules, picked off one by one:

Module	Before → After	Module	Before → After
api_token (stable-token/rotate)	0 → 90%	context_builder	14 → 52%
retrieval_packets	13 → 81%	memory_reset (privacy wipe)	11 → 44%
operator_goal_actions	14 → 78%	operator_actions (proposal queue)	16 → 44%
tool_result_normalizer	18 → 74%	approval_engine (autonomy gate)	19 → 34%
grounded_control (contract)	15 → 61%	personal_memory_surface ¹	6 → 10%
tool_execution_authority	16 → 59%	deterministic_grounding_gate ²	24 → 34%

¹ Only the pure `is_personal_memory_query` classifier is reachable; the rest builds a live surface from the memory DB (not testable headless) — reported honestly, not padded.

This pass also **found and fixed a latent bug**: `canonicalize_execution_call` advertised tool/name aliases for action, but they were dead code — a `{"tool": "X"}` call resolved to `action=""` because `_payload_from_packet_like` returned the plain dict first. Now fixed and tested.

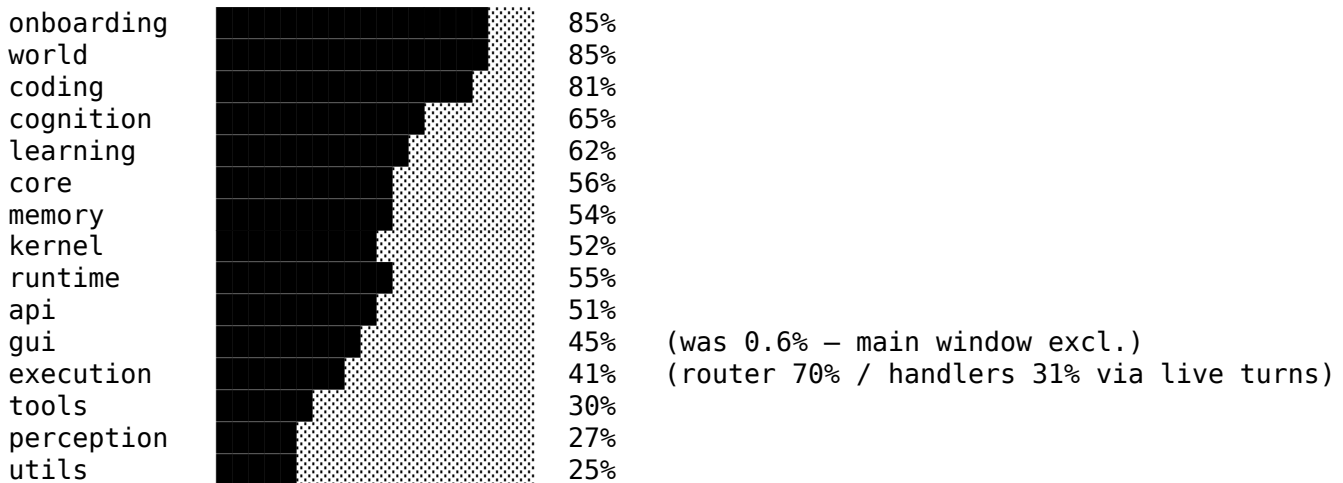
Bias throughout: the **security/privacy-critical** paths an auditor checks first — bearer/RBAC auth, the fail-closed shell allowlist, the hardcoded-path/PII scanners, the anti-confabulation guards, and DB-path isolation.

4. Coverage by subsystem

Subsystem	Cover	Stmts	Subsystem	Cover	Stmts
eli/onboarding	85%	159	eli/gui	45%	5,439
eli/world	85%	971	eli/contracts	61%	256
eli/coding	81%	1,045	eli/planning	45%	2,132
eli/cognition	65%	7,021	eli/execution	41%	12,309
eli/learning	62%	1,520	eli/plugins	39%	1,050
eli/core	56%	3,492	eli/system	37%	261

Subsystem	Cover	Stmts	Subsystem	Cover	Stmts
eli/memory	54%	3,338	eli/tools	30%	4,508
eli/kernel	52%	7,059	eli/perception	27%	4,138
eli/runtime	55%	13,131	eli/integrations	26%	507
api (web)	51%	1,204	eli/utils	25%	468
			(gui main window)	excl.	~7k

Visual (testable surface)



5. GUI recovery (the offscreen-Qt effort)

The GUI was 0.6%. The **main window** (eli_pro_audio_gui_MKI.py, ~7k stmts) blocks on device/display init and can't be built in CI — that stays excluded. Everything else constructs standalone under QT_QPA_PLATFORM=offscreen, so lane 4 builds them all for real (27 tests):

GUI file	Cover	GUI file	Cover
tabs/eli_world_tab.py	100%	docks/operator_console_dock.py	67%
docks/proactive_dock.py	94%	panels/settings.py	60%
tabs/experimental_tab.py	82%	widgets/ollama_model_selector.py	50%
panels/startup.py	61%	coding_tab.py	49%
labs_tab.py	40%	tabs/tasks_tab.py	45%

LabsTab builds its full 400+-widget tree; the settings dialog builds 268 children; the startup wizards, both docks, and every tab construct and wire their UI. The only GUI left uncovered is the main window itself (excluded) and residual event-handler branches that need real user interaction.

6. Why the remaining low subsystems are low

The low scores cluster around **one cause: the I/O boundaries where ELI touches the real world.** Pure logic is well covered; the edges aren't.

Subsystem	Why it's low
execution (41%)	~174 action handlers, most side-effecting (open apps, shell, screenshots, media). Can't unit-test "open Firefox" — the test <i>does it</i> . Covered via safe live-lane turns. Router beside it is 70% .
tools (30%)	News fetcher (network, gated off), image engine (GPU diffusion), weather (network). Non-network logic is tested; fetch/GPU paths aren't.
perception (27%)	GPU vision, mic STT, TTS, gaze, live-desktop control — none runs headless . Covered part = the pure parsers (equations 100%, CSV 78%).
utils (25%)	Mostly platform_compat.py — if WINDOWS ... elif MACOS ...; on Linux CI only the Linux branch runs. Inherent to cross-platform code.

7. The 5 pre-existing reds (none from this work)

1-3. smart_home plugin — the in-progress Home-Assistant removal (voice SMART_HOME now uses ELI's own MQTT server). 4. A blueprint references a since-moved handlers module (the handlers package `__init__.py`). 5. Silent-swallow ratchet — 987 except: pass vs a 950 ceiling (a standing observability debt; the ratchet test correctly forbids raising the ceiling).

8. On comparisons (a "closest-in-spirit" project claiming ~89%)

The gap is largely *what ELI is*: a large **embodiment surface** — desktop GUI, gaze/webcam, mic/voice, local vision, OS control, smart-home — that can't fully run in headless CI. A leaner pure-software agent lacks that surface, so a higher fraction of its code is unit-testable by construction. ELI's cognitive **core is in a comparable band** (coding 81%, cognition 65%, kernel 52%), and this report **honestly counts the whole surface** — including the GUI (now 45%) — rather than hiding the hard parts. The residual gap is the genuinely un-automatable edges: side-effecting OS actions, physical hardware, and the one window that can't be built headless.

9. Reproduce

```
bash scripts/coverage_full.sh          # runs all 4 lanes, combines, reports
# outputs: artifacts/coverage_report.txt + artifacts/coverage_html/index.html
```

Config: `.coveragerc` omits only the headless-untestable main window. Every number in this report regenerates from that command on any checkout.